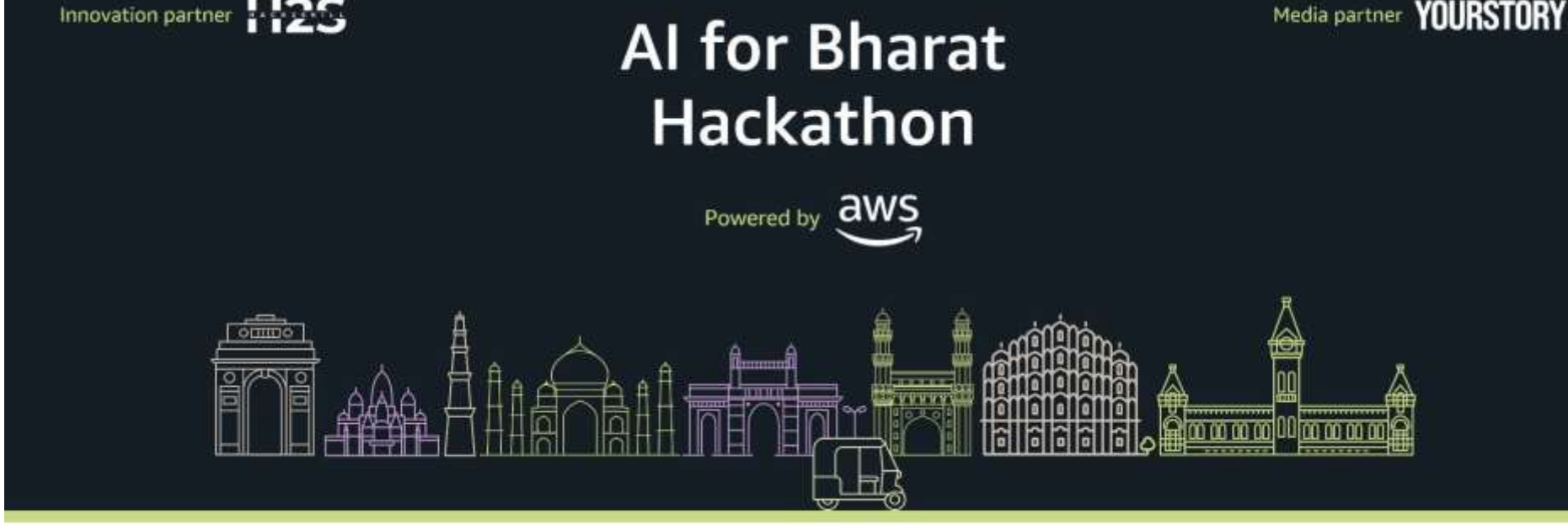


Team Name : AeroNova-Drishti

Team Leader Name : Dhrumi Modi

Problem Statement : Improving real-time disaster response coordination using AI & aerial intelligence.

Brief about the Idea :

AeroNova – Drishti is an AI-powered disaster management platform that uses:
Drones (Aerial Surveillance)
Satellite Data
AI-based Damage Detection
Real-time Coordination Dashboard
It provides:
Live disaster mapping
Victim detection
Resource allocation optimization
Multi-agency coordination
Goal: Faster, Smarter, Coordinated Disaster Response

Your solution should be able to explain the following:

- **How different is it from any of the other existing ideas?**

Existing Systems :
- Manual reporting
- Delayed updates
- Separate systems
- Limited prediction

AeroNova-Drishti:
- AI-based automatic detection
- Real-time live drone feeds
- Unified AI dashboard
- Predictive risk analysis

How Will It Solve the Problem?

Drones capture live disaster visuals
AI detects:
Trapped victims
Flood level
Collapsed buildings
System prioritizes rescue zones
Dashboard assigns teams automatically
Authorities receive alerts instantly
➞ Reduces response time
➞ Improves coordination
➞ Saves lives

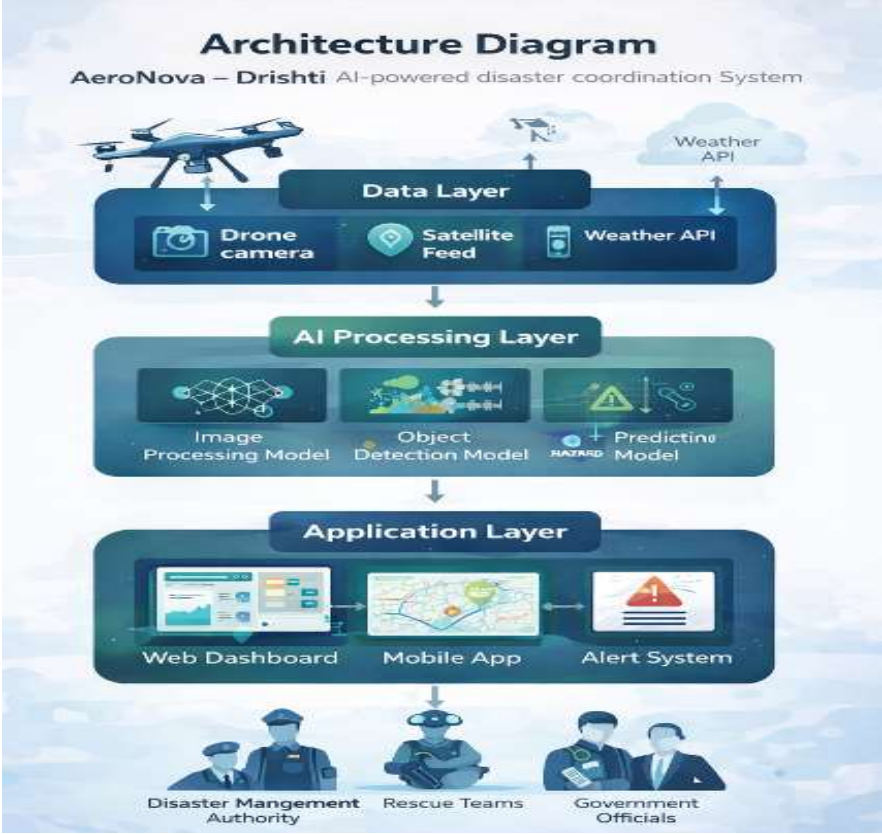
USP of the proposed solution

- ✓ AI-powered victim detection
- ✓ Smart resource allocation
- ✓ Real-time disaster heat maps
- ✓ Predictive disaster risk analysis
- ✓ Works even in low-network zones

List of features offered by the solution:

Process flow diagram or Use-case diagram

Architecture diagram of the proposed solution:



Technologies to be used in the solution:

Python
TensorFlow / PyTorch
OpenCV
AWS Cloud
AWS S3 (Storage)
AWS Lambda
GIS Mapping Tools
React / Flutter (Frontend)
Firebase / MongoDB (Database)

Estimated implementation cost (optional):

Component
Estimated Cost
AI Model Development
₹3–5 Lakhs
Drone Hardware
₹2–4 Lakhs
Cloud Infrastructure
₹1–2 Lakhs/year
App & Dashboard
₹2 Lakhs
Total Estimated
Cost: ₹8–12 Lakhs (Scalable Model)

Impact in India:

Faster disaster response

Reduced human casualties

Better coordination

Useful in rural & urban areas

Scalable for National Disaster Management Authority (NDMA)

Conclusion:

AeroNova – Drishti combines:

AI + Drones + Real-Time Intelligence

To create a Smarter, Faster & Safer Disaster Response System for Bharat

Innovation partner

Media Partner YOURSTORY

AI for Bharat Hackathon

Powered by aws

Thank You